

RANK CORRELATION

Spearman's rank correlation coefficient is defined as:

$$R = 1 - \frac{6 \sum D^2}{N(N^2 - 1)} \text{ (OR) } R = 1 - \frac{6 \sum D^2}{N^3 - N}$$

1. The ranking of 10 students in two subjects A and B are as follows

A	6	5	3	10	2	4	9	7	8	1
B	3	8	4	9	1	6	10	7	5	2

Find the spearman's rank correlation coefficient.

R1	R2	(R1-R2)²=D²
6	3	9
5	8	9
3	4	1
10	9	1
2	1	1
4	6	4
9	10	1
7	7	0
8	5	9
1	2	1
$\sum D^2 = 36$		

$$R = 1 - \frac{6 \sum D^2}{N^3 - N} = 1 - \frac{6 \times 36}{10^3 - 10} = 1 - \frac{216}{990} = 0.782$$

2. Practice problem:

Find Spearman's rank correlation coefficient.

X	1	6	5	10	3	2	4	9	7	8
Y	3	5	8	4	7	10	2	1	6	9
Z	6	4	9	8	1	2	3	10	5	7

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3. Calculate spearman's coefficient of rank correlation for the following:

X	53	98	95	81	75	61	59	55
Y	47	25	32	37	30	40	39	45

X	Y	Rank X (R1)	Rank Y (R2)	R1-R2=D	D ²
53	47	8	1	7	49
98	25	1	8	-7	49
95	32	2	6	-4	16
81	37	3	5	-2	4
75	30	4	7	-3	9
61	40	5	3	2	4
59	39	6	4	2	4
55	45	7	2	5	25
					$\sum D^2 = 160$

$$R = 1 - \frac{6 \sum D^2}{N^3 - N} = 1 - \frac{6 \times 160}{8^3 - 8} = 1 - 1.9048 = -0.905$$

4. From the following data of the ranks obtained by 8 students in the computer science and statistics papers, Compute rank correlation coefficient.

Marks in computer science	15	20	28	12	40	60	20	80
Marks in statistics	40	30	50	30	20	10	30	60

Marks in computer science(X)	Marks in statistics(Y)	RankX (R1)	RankY (R2)	D=R1-R2	D ²
15	40	7	3	4	16
20	30	5.5	5	0.5	0.25
28	50	4	2	2	4
12	30	8	5	3	9
40	20	3	7	-4	16
60	10	2	8	-6	36
20	30	5.5	5	0.5	0.25
80	60	1	1	0	0

	$\sum D^2 = 81.5$
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$$R = 1 - \frac{6 \sum D^2}{N^3 - N} = 0.0298$$

Regression lines(or)Linear Regression

The equation of the line of regression of X on Y is given by

$$(X - \bar{x}) = r \frac{\sigma_x}{\sigma_y} (Y - \bar{y})$$

and the equation of line of regression of Y on X is given by

$$(Y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (X - \bar{x})$$

Where 'r' is the correlation coefficient between X and Y. σ_x and σ_y represents the standard deviation of X and Y respectively.

The quantities $r \left(\frac{\sigma_x}{\sigma_y} \right)$ and $r \left(\frac{\sigma_y}{\sigma_x} \right)$ are called regression coefficients. The product of regression coefficients is equal to r^2 .

Problems:

1. Find the mean values of the variables X and Y and correlation coefficient from the following regression equations.

$$2Y - X = 50 \dots\dots\dots(1)$$

$$3Y - 2X = 10 \dots\dots\dots(2)$$

Solve the equations (1) and (2)

$$X=130=\bar{X} ; Y=90=\bar{Y}$$

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Rewriting (1) and (2) as

$$Y = \frac{1}{2}X + 25; \quad [(Y - \bar{Y}) = r \frac{\sigma_y}{\sigma_x}(X - \bar{X})]$$

$$X = \frac{3}{2}Y - 5; \quad [(X - \bar{X}) = r \frac{\sigma_x}{\sigma_y}(Y - \bar{Y})]$$

$$r \frac{\sigma_y}{\sigma_x} = 1/2; \quad r \frac{\sigma_x}{\sigma_y} = 3/2$$

$$r^2 = \frac{3}{4} \therefore r = 0.866$$

2. The correlation coefficient between two random variables X and Y is $r=0.6$ if $\sigma_x=1.5, \sigma_y=2, \bar{X}=10$ and $\bar{Y}=20$. Find the regression lines of (i)Y on X and (ii)X on Y.

Regression lines of X on Y is $(X - \bar{x}) = r \frac{\sigma_x}{\sigma_y}(Y - \bar{y})$

$$(X - 10) = 0.6\left(\frac{1.5}{2}\right)(Y - 20) \Rightarrow X = 0.45Y + 1$$

Regression lines of Y on X is $(Y - \bar{y}) = r \frac{\sigma_y}{\sigma_x}(X - \bar{x})$

$$(Y - 20) = 0.6\left(\frac{2}{1.5}\right)(X - 10) \Rightarrow Y = 0.8X + 12$$

3. From the following data , obtain two regression equations

X	6	2	10	4	8
Y	9	11	5	8	7

X	$x = X - \bar{x}$	x^2	Y	$y = Y - \bar{y}$	y^2	xy
6	0	0	9	1	1	0
2	-4	16	11	3	9	-12
10	+4	16	5	-3	9	-12
4	-2	4	8	0	0	0
8	+2	4	7	-1	1	-2
$\sum X = 30$	$\sum x = 0$	$\sum x^2 = 40$	$\sum Y = 40$	$\sum y = 0$	$\sum y^2 = 20$	$\sum xy = -26$

$$\bar{x} = \frac{\sum X}{N} = \frac{6+2+10+4+8}{5} = \frac{30}{5} = 6$$

$$\bar{y} = \frac{\sum Y}{N} = \frac{40}{5} = 8$$

The equation of the line of regression of X on Y is given by

$$(X - \bar{x}) = r \frac{\sigma_x}{\sigma_y} (Y - \bar{y})$$

$$r \frac{\sigma_x}{\sigma_y} = \frac{\sum xy}{\sum y^2} = \frac{-26}{20} = -1.3$$

$$\therefore X - 6 = -1.3(Y - 8) \Rightarrow X = 16.4 - 1.3Y$$

The equation of line of regression of Y on X is given by

$$(Y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (X - \bar{x})$$

$$r \frac{\sigma_y}{\sigma_x} = \frac{\sum xy}{\sum x^2} = \frac{-26}{40} = -0.65$$

$$Y - 8 = -0.65(X - 6) \Rightarrow Y = 11.9 - 0.65X$$

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4. In a partially destroyed record, the following data and legible:
Variance of $X=25$. Regression equation of X on Y is $5X-Y=22$ and
regression equation of Y on X is $64X-45Y=24$.
Find 1) Mean values of X and Y 2) Coefficient of correlation between
 X and Y and (3) Standard deviation of Y .

Ans:1) $X = \bar{x} = 6$; $Y = \bar{y} = 8$

Rewrite equation 1 and 2

$$2) r = 8/15 \quad (\because \left(r \frac{\sigma_x}{\sigma_y} \right) \left(r \frac{\sigma_y}{\sigma_x} \right) = r^2)$$

$$3) \sigma_y = 13.33(\text{or}) = \frac{40}{3} \quad (\because r \frac{\sigma_x}{\sigma_y} = \frac{1}{5} \Rightarrow \sigma_y = 5.r.\sigma_x)$$

$$(\text{given } V(X)=25 \longrightarrow \sigma_x = \sqrt{V(X)} = \sqrt{25} = 5)$$

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